TP- investigate static routes

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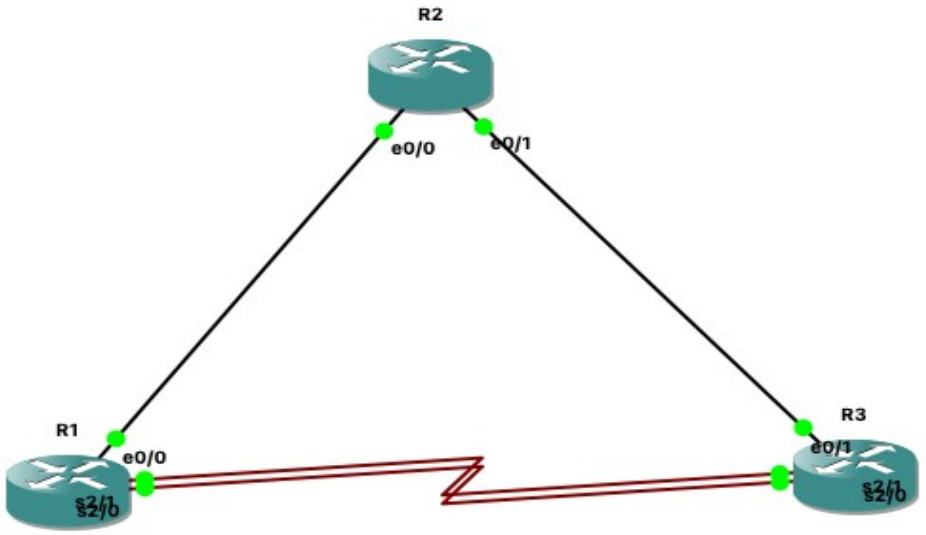
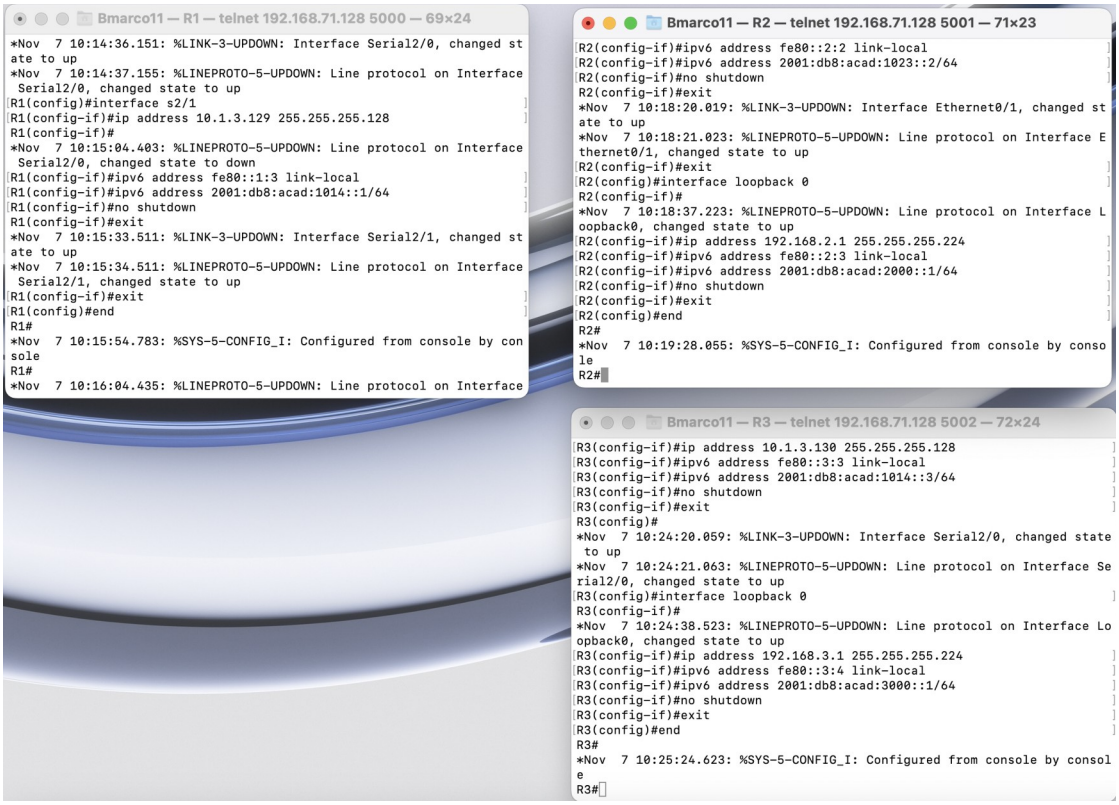
Topology





Bmarco11 — R1 — telnet 192.168.71.128 5000 — 69×24

```
*Nov  7 10:14:36.151: %LINK-3-UPDOWN: Interface Serial2/0, changed st
ate to up
*Nov  7 10:14:37.155: %LINEPROTO-5-UPDOWN: Line protocol on Interface
 Serial2/0, changed state to up
R1(config)#interface s2/1
R1(config-if)#ip address 10.1.3.129 255.255.255.128
R1(config-if)#
*Nov  7 10:15:04.403: %LINEPROTO-5-UPDOWN: Line protocol on Interface
 Serial2/0, changed state to down
R1(config-if)#ipv6 address fe80::1:3 link-local
R1(config-if)#ipv6 address 2001:db8:acad:1014::1/64
R1(config-if)#no shutdown
R1(config-if)#exit
*Nov  7 10:15:33.511: %LINK-3-UPDOWN: Interface Serial2/1, changed st
ate to up
*Nov  7 10:15:34.511: %LINEPROTO-5-UPDOWN: Line protocol on Interface
 Serial2/1, changed state to up
R1(config-if)#exit
R1(config)#end
R1#
*Nov  7 10:15:54.783: %SYS-5-CONFIG_I: Configured from console by con
sole
R1#
*Nov  7 10:16:04.435: %LINEPROTO-5-UPDOWN: Line protocol on Interface
```

Bmarco11 — R2 — telnet 192.168.71.128 5001 — 71×23

```
R2(config-if)#ipv6 address fe80::2:2 link-local
R2(config-if)#ipv6 address 2001:db8:acad:1023::2/64
R2(config-if)#no shutdown
R2(config-if)#exit
*Nov  7 10:18:20.019: %LINK-3-UPDOWN: Interface Ethernet0/1, changed st
ate to up
*Nov  7 10:18:21.023: %LINEPROTO-5-UPDOWN: Line protocol on Interface E
thernet0/1, changed state to up
R2(config-if)#exit
R2(config)#interface loopback 0
R2(config-if)#
*Nov  7 10:18:37.223: %LINEPROTO-5-UPDOWN: Line protocol on Interface L
oopback0, changed state to up
R2(config-if)#ip address 192.168.2.1 255.255.255.224
R2(config-if)#ipv6 address fe80::2:3 link-local
R2(config-if)#ipv6 address 2001:db8:acad:2000::1/64
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#end
R2#
*Nov  7 10:19:28.055: %SYS-5-CONFIG_I: Configured from console by conso
le
R2#
```

Bmarco11 — R3 — telnet 192.168.71.128 5002 — 72×24

```
R3(config-if)#ip address 10.1.3.130 255.255.255.128
R3(config-if)#ipv6 address fe80::3:3 link-local
R3(config-if)#ipv6 address 2001:db8:acad:1014::3/64
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Nov  7 10:24:20.059: %LINK-3-UPDOWN: Interface Serial2/0, changed state
 to up
*Nov  7 10:24:21.063: %LINEPROTO-5-UPDOWN: Line protocol on Interface Se
rial2/0, changed state to up
R3(config)#interface loopback 0
R3(config-if)#
*Nov  7 10:24:38.523: %LINEPROTO-5-UPDOWN: Line protocol on Interface Lo
opback0, changed state to up
R3(config-if)#ip address 192.168.3.1 255.255.255.224
R3(config-if)#ipv6 address fe80::3:4 link-local
R3(config-if)#ipv6 address 2001:db8:acad:3000::1/64
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#end
R3#
*Nov  7 10:25:24.623: %SYS-5-CONFIG_I: Configured from console by consol
e
R3#
```

Bmarco11 — R1 — telnet 192.168.71.128 5000 — 77×24

```
R1#
*Nov  7 10:16:04.435: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2
/1, changed state to down
R1#
*Nov  7 10:24:24.635: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2
/0, changed state to up
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 192.168.2.0 255.255.255.224 e0/0
R1(config)#end
R1#
*Nov  7 10:30:32.931: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route static | begin Gateway
Gateway of last resort is not set

      192.168.2.0/27 is subnetted, 1 subnets
S        192.168.2.0 is directly connected, Ethernet0/0
R1#show ip route 192.168.2.0 255.255.255.224
Routing entry for 192.168.2.0/27
  Known via "static", distance 1, metric 0 (connected)
  Routing Descriptor Blocks:
  * directly connected, via Ethernet0/0
      Route metric is 0, traffic share count is 1
R1#
```

Bmarco11 — R1 — telnet 192.168.71.128 5000 — 77×24

```
Gateway of last resort is not set

      192.168.2.0/27 is subnetted, 1 subnets
S        192.168.2.0 is directly connected, Ethernet0/0
R1#show ip route 192.168.2.0 255.255.255.224
Routing entry for 192.168.2.0/27
  Known via "static", distance 1, metric 0 (connected)
  Routing Descriptor Blocks:
  * directly connected, via Ethernet0/0
      Route metric is 0, traffic share count is 1
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 10.2.3.0 255.255.255.0 10.1.2.2
R1(config)#end
R1#
*Nov  7 10:33:03.691: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route static | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
S        10.2.3.0/24 [1/0] via 10.1.2.2
      192.168.2.0/27 is subnetted, 1 subnets
S        192.168.2.0 is directly connected, Ethernet0/0
R1#
```

Bmarco11 — R1 — telnet 192.168.71.128 5000 — 77×23

```
      10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
S        10.2.3.0/24 [1/0] via 10.1.2.2, Ethernet0/0
      192.168.2.0/27 is subnetted, 1 subnets
S        192.168.2.0 is directly connected, Ethernet0/0
      192.168.3.0/27 is subnetted, 1 subnets
S        192.168.3.0 [1/0] via 10.1.3.130, Serial2/0
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 192.168.3.0 255.255.255.224 s2/1 10.1.3.3
R1(config)#end
R1#
*Nov  7 10:56:04.311: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route static | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
S        10.2.3.0/24 [1/0] via 10.1.2.2, Ethernet0/0
      192.168.2.0/27 is subnetted, 1 subnets
S        192.168.2.0 is directly connected, Ethernet0/0
      192.168.3.0/27 is subnetted, 1 subnets
S        192.168.3.0 [1/0] via 10.1.3.130, Serial2/0
R1#
```

Bmarco11 — R1 — telnet 192.168.71.128 5000 — 77×29

```
R1(config)#end
R1#
*Nov  7 10:39:35.135: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route static | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
S        10.2.3.0/24 [1/0] via 10.1.2.2, Ethernet0/0
      192.168.2.0/27 is subnetted, 1 subnets
S        192.168.2.0 is directly connected, Ethernet0/0
R1#show ip route 10.2.3.0 255.255.255.0
Routing entry for 10.2.3.0/24
  Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
  * 10.1.2.2, via Ethernet0/0
      Route metric is 0, traffic share count is 1
R1#show ip route 10.1.2.1
Routing entry for 10.1.2.1/32
  Known via "connected", distance 0, metric 0 (connected)
  Routing Descriptor Blocks:
  * directly connected, via Ethernet0/0
      Route metric is 0, traffic share count is 1
R1#show ip route 10.1.2.0 255.255.255.0
Routing entry for 10.1.2.0/24
  Known via "connected", distance 0, metric 0 (connected, via interface)
  Routing Descriptor Blocks:
  * directly connected, via Ethernet0/0
      Route metric is 0, traffic share count is 1
R1#
```

Bmarco11 — R3 — telnet 192.168.71.128 5002 — 72×24

```
R3#
*Nov  7 11:07:14.791: %SYS-5-CONFIG_I: Configured from console by consol
e
R3#show ipv6 route static
IPv6 Routing Table - default - 6 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP ext
ernal
       ND - Neighbor Discovery
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext
 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
S   3333::/64 [1/0]
     via Null0, directly connected
R3#show ipv6 route 3333::/64
Routing entry for 3333::/64
  Known via "static", distance 1, metric 0
  Route count is 1/1, share count 0
  Routing paths:
    directly connected via Null0
      Last updated 00:04:03 ago

R3#
```

Bmarco11 — R1 — telnet 192.168.71.128 5000 — 77×23

```
R1#show ip route 192.168.3.0 255.255.255.224
Routing entry for 192.168.3.0/27
  Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
  * 10.1.3.130, via Serial2/0
      Route metric is 0, traffic share count is 1
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no ip route 192.168.3.0 255.255.255.224 s2/0 10.1.3.130
R1(config)#ip route 192.168.3.0 255.255.255.224 s2/0 10.1.3.130 7
R1(config)#end
R1#
*Nov  7 11:00:40.611: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route static | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
S        10.2.3.0/24 [1/0] via 10.1.2.2, Ethernet0/0
      192.168.2.0/27 is subnetted, 1 subnets
S        192.168.2.0 is directly connected, Ethernet0/0
      192.168.3.0/27 is subnetted, 1 subnets
S        192.168.3.0 [7/0] via 10.1.3.130, Serial2/0
R1#
```

Bmarco11 — R1 — telnet 192.168.71.128 5000 — 77×23

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 209.165.200.0 255.255.255.0 null 0
R1(config)#end
R1#
*Nov  7 11:17:50.887: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route static | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
S        10.2.3.0/24 [1/0] via 10.1.2.2, Ethernet0/0
      192.168.2.0/27 is subnetted, 1 subnets
S        192.168.2.0 is directly connected, Ethernet0/0
S     209.165.200.0/24 is directly connected, Null0
R1#show ip route 209.165.200.0 255.255.255.0
Routing entry for 209.165.200.0/24
  Known via "static", distance 1, metric 0 (connected)
  Routing Descriptor Blocks:
  * directly connected, via Null0
      Route metric is 0, traffic share count is 1
R1#
R1#
```

Bmarco11 — R3 — telnet 192.168.71.128 5002 — 72×24

```
e
R3#show ipv6 route static
IPv6 Routing Table - default - 7 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP ext
ernal
       ND - Neighbor Discovery
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext
 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
S   2001:DB8:ACAD:1012::/64 [1/0]
     via FE80::2:2, Ethernet0/1
S   3333::/64 [1/0]
     via Null0, directly connected
R3#show ipv6 route 2001:db8:acad:1012::64
Routing entry for 2001:DB8:ACAD:1012::/64
  Known via "static", distance 1, metric 0
  Route count is 1/1, share count 0
  Routing paths:
    FE80::2:2, Ethernet0/1
      Last updated 00:00:39 ago

R3#
```